SPE-227257-MS

Conservation of Fresh Water with Groundbreaking Substitutes

Ataur R. Malik, Majed H. Daghriri, Hamad F. Kulaib, and Abdullah A. Ghamdi, Saudi Aramco, Saudi Arabia

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Ground water from water supply wells is ordinarily used as the base fluid for well stimulation. The water consumption per stage during well stimulation can be significant. Preserving the groundwater resources by providing its substitute to carry out major upstream operations is perceived with a great priority. The objective of this paper is to demonstrate several ground water conservation schemes that were proven at the field during hydraulic fracturing treatment.

Primarily, the water quantity and quality of evaporation pond at the gas plant has been evaluated for re-use opportunity for hydraulic fracturing operations with favorable geochemical composition. The boiler blowdown, produced, brackish and storm water are collected at the pond. Secondly, a novel technology was introduced for distilled produced water which is purely a water management technology aiming to desalinate produced water by transforming its ionic properties to deliver a high value end-product. Finally, seawater was nano-filtered to reduce sulphate content from 4000+ ppm to less than 500 ppm to use as base for stimulation fluids.

Once the geo-chemical analysis of one of the alternative water sources is found satisfactory, various types of fracturing fluids (i.e., zirconate crosslinked gel, visco-elastic surfactant based fracturing fluid system, various acid blends etc.) were formulated. Intensive rheology, scaling and core flooding tests were performed on theses fluid systems to assess the fluids stability, compatibility and the formation damage level at the reservoir conditions. The ground water substitute-based stimulation fluids sustained viscosity values above the minimum requirement of 300 centipoise at 100 1/s shear rate for longer than the pumping time and then broke into the level of linear gel. The core flooding results showed high retained permeability indicating minimal formation damage. These stimulation fluid systems have been successfully trial tested in the high temperature carbonate reservoirs replacing ground water by 100%. The acid blends formulated with ground water substitutes and standard corrosion inhibition package for high temperature didn't show any signs of incompatibility such as fluid phase separation, flocculation, or precipitation at the reservoir conditions.

The ground water substitutes are measured as resources instead of waste streams. The distilled produced water technology has a custom designed dynamic vapor recovery system that is capable of removing salt from hypersaline oilfield produced water. Recycling, retreating and re-using them with careful evaluations

of chemistry and recipes allowed to manage well stimulation activities in a much better environmentally friendly way without compromising the stimulation effectiveness and well performance.

Introduction

Ground water utilization from water supply wells continues to dominate the base fluid used across rigless intervention and stimulation in the region given its compatibility stand. For example, the average consumption per stage during gas well stimulation is nearly 2,000+ bbl. Over the years, enormous efforts are put to optimize water consumption during frac operations by certifying a family of chemicals and completion solutions that serve to increase the effectiveness of stimulation work through acid retardation, chemicals diversion, and reduced intervention scope and subsequently optimized water consumption per stage to lesser values. Over the years, several ground water conservation projects were field implemented as per production engineering design and programs in collaboration with all involved stakeholders. These projects were executed to affirm the technical feasibility of utilizing treated recycled and produced water that exhibited similar geochemical properties to that of water originated from water supply wells currently being in use across the rigless operations. The multidisciplinary technical team has collaborated with inhouse and Service Providers' laboratories to develop the proper fluid formulations with required additives added to the treated water to ensure that the stimulation fluid blends exhibit desired characteristics in terms of thermal stability, rheological properties and reservoir compatibility. Brief summary on water conservation projects (distilled produced water, evaporation pond water and nano-filtered seawater) that were field implemented during rigless intervention and stimulation activities in the gas and oil wells are presented in this paper.

Well Features

Well A, B and C are the candidate wells where different fresh water substitute-based stimulation fluids have been used. Well A is a single lateral well that was horizontally geo-steered across carbonate reservoir achieving 4,000+ ft of gross thickness. The open hole logs showed 40%+ net-to-gross ratio with moderate average porosity. Well B was drilled as a vertical well across two carbonate intervals. Well C was drilled as a vertical S-shape well across two carbonate intervals.

Technology Description and Field Implementation

Distilled Produced Water (DPW)

The DPW is purely a water management technology that aims to desalinate oil field produced water by transforming its ionic properties to deliver a high value end-product. It has a custom designed dynamic vapor recovery system that is capable of removing salt from hypersaline oil field produced water. The DPW distillate is comparable to fresh water and can serve multiple oilfield applications including water injection for reservoir support, hydraulic fracturing, Coiled Tubing (CT) well interventions and wash water for crude oil desalination etc. The DPW was successfully pumped in two stages of Well B conserving 1,000 bbl. of fresh ground water. All acid blends and gel systems were mixed using DPW water during the acid fracturing treatments.

Well A. Well A was drilled as a horizontal gas producer across the lower carbonate formation and completed as open hole MSF. This high pressure and high temperature well was acid stimulated in two stages with DPW based stimulation fluids. Well A was selected as the candidate for DPW based stimulation fluids for the following reasons:

- Satisfactory and comparable geo-chem analysis to the ground water.

- DPW based PAD fluid developed satisfactory rheology profile at reservoir conditions using multiple shear rates.
- DPW based stimulation fluid was formulated with an appropriate scale inhibitor that exhibited no precipitation of scale.

Stimulation Fluids Description. Acid fracturing treatment in carbonate formations included alternating crosslinked gel (PAD), regular HCl (26%), emulsified acid (26%) and diverter. The PAD fluid used to create the fracture profile and fingering effect is zirconate crosslinked gel. The acid volume during the stimulation treatment was around 250 gal per feet of net pay.

The laboratory tests were performed focusing on:

- Stability and viscosity breaking of zirconate crosslinked gel at 305°F.
- Stability at room temperature of 26% HCl straight acid, emulsified acid and Visco-Elastic diverted acid.
- Rheological performance of spent diverter at 250°F.

Laboratory Results. The DPW QA/QC and chemical composition are shown in Table-1 with acceptable results.

Table 1—QA/QC on DPW water sample

Variable	Filtered recycle water
pH	7.9
Specific Gravity	1.001
Calcium, ppm	30
Magnesium, ppm	14
Bicarbonates, ppm	26
Iron, ppm	0.5
Sulfates, ppm	10
Chloride's ppm	50
TDS ppm	163

Fig-1 shows the viscosity as function of time at BH temperature. Chandler test conducted to DPW crosslinked gel which resulted in a stabilized viscosity profile (>300 cP) and favorable breaking behavior (<100 cP), high pressure high temperature (HPHT) conditions (305 F) using multiple shear rates.

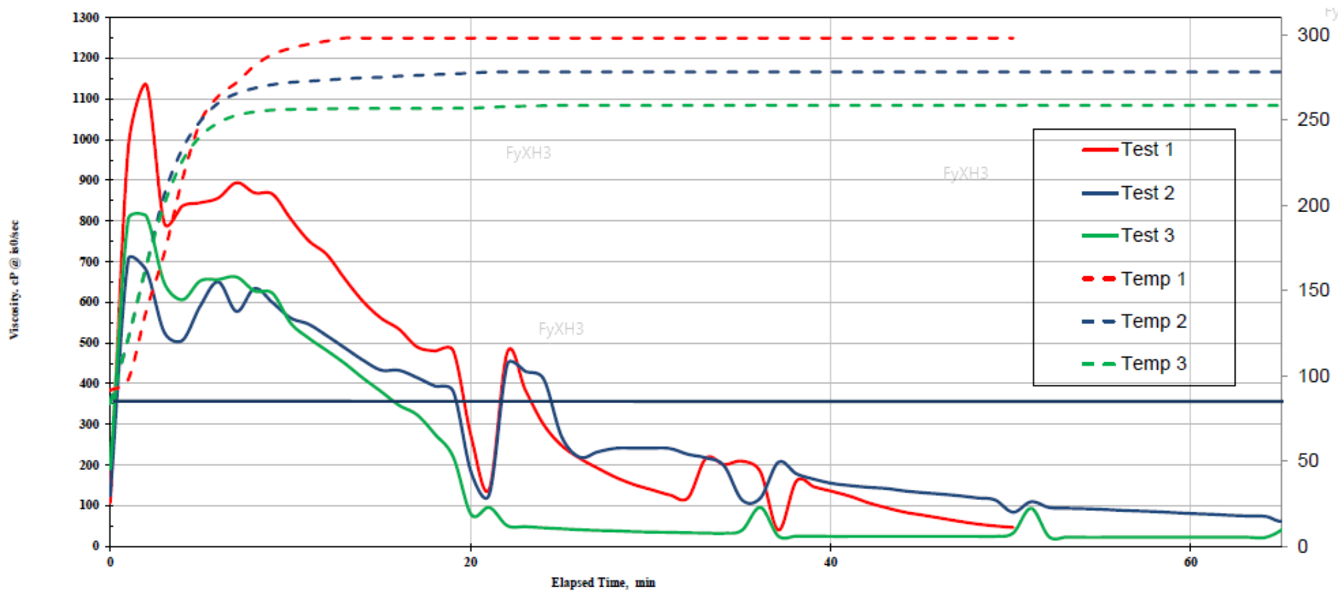

Figure 1—Rheology of crosslinked gel at BH temperatures

Intensive lab tests were conducted to test the stability of DPW based stimulation fluid systems. As can be seen in Fig-2 and Fig-3, the DPW is compatible with the various stimulation fluid systems indicating no separation or precipitation in the fluids. Also, the drop test/ snake test showed no dispersing out of the emulsified acid.

Figure 2—Stability of regular and emulsified acid system at room temperature

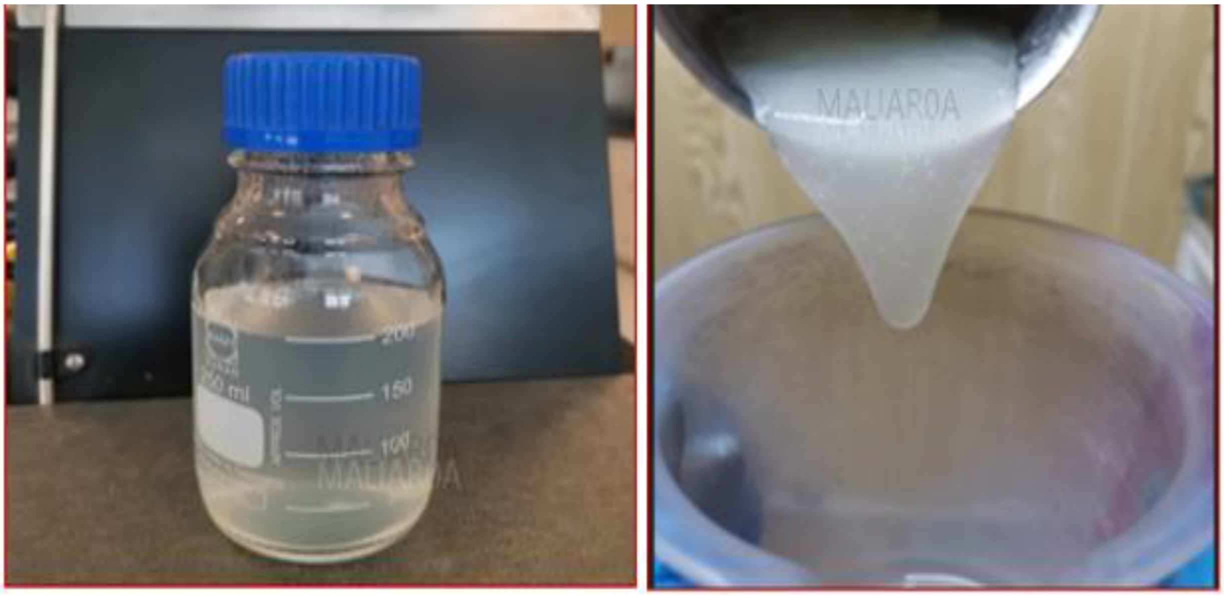

Figure 3—Gel stability test

Stimulation Design and Execution. Design: The design includes hydrochloric acid (26% HCl) and emulsified acid (26% HCl based) as main reactive treatment fluid using an average combined volume 250 gal per ft. of perforation. The zirconate based crosslinked gel as Pad was optimized with DPW. The Acid to PAD ratio was selected to be 8:3 based on reservoir petrophysical and lithology properties. To minimize fluid leak off to high permeable zones and improve diversion effects, diverter was included in the stimulation design.

Execution (Lower stage / stage-1):

Initial WH pressure was 5,000 psi. Stage-1 injection test (Fig-4) was started by pumping treated water at 1.0 bpm and then rate was increased to maximum of 9 bpm. Total volume of DPW pumped was 130.0 bbl. The maximum surface treatment pressure (STP) was 12,374 psi and maximum bottomhole treatment pressure (BHTP) was 18,040 psi. The DPW trial test was performed for Well-A (Fig- 5) Lower Carbonate formation with an initial pumping rate of 3.0 bpm that exhibited stable surface and bottomhole treatment pressures. After initial acid volumes reached formation, pumping rate was increased to maximum of 26 bpm. Total fluid pumped was 2,013 bbl. Acid was flushed and over flushed with 387 bbl of treated DPW. Maximum STP was 12,003 psi while the maximum BHTP was 17,955 psi.

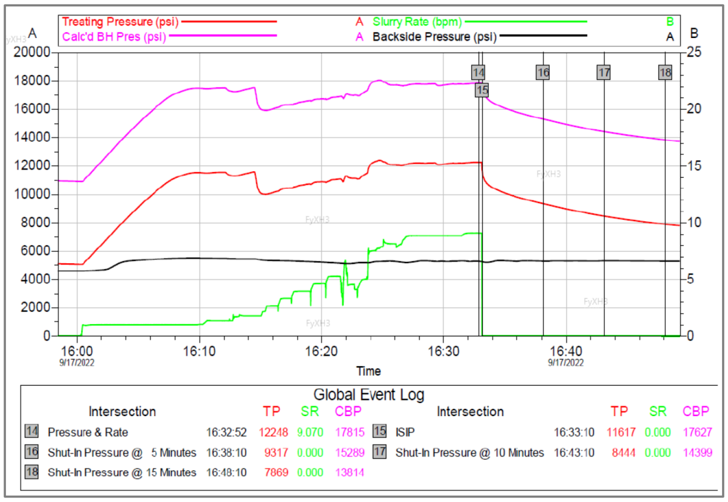

Figure 4—Injection Test for stage-1 (Well A - lower stage / stage-1)

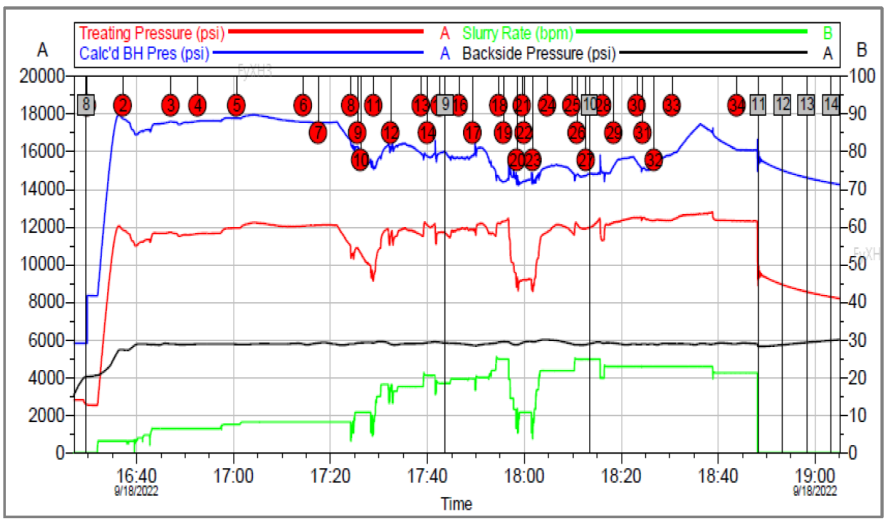

Figure 5—Acid stimulation for stage-1 (Well A - lower stage / stage-1)

Table-2 shows the actual pumping schedule. The actual pumped stimulation fluids volume (Table-3) was very close to the treatment design.

Table 2—Stage-1 pumping schedule (Well-A lower stage / stage-1)

Stage Number	Description	Fluid System	Clean Volume gal	Slurry Volume gal	Avg Slurry Rate bpm	Avg Treating Pressure psi	Max Slurry Rate bpm	Max Treat Pr psi
1	Spacer	Treated Water	500	500	3.1	8763.99	3.2	12047.36
2	Acid	26% Gelled Acid	2000	2000	4.8	11468.20	6.6	11830.27
3	Spacer	Treated Water	1500	1500	6.4	11591.07	6.5	11701.54
4	Pad	30# x-linked Gel	2300	2333	6.9	11747.55	7.5	11935.22
5	Acid	26% Emulsified Acid	4580	4580	8.2	12083.09	8.2	12221.27
6	Pad	30# x-linked Gel	1000	1014	8.2	12065.05	8.2	12084.66
7	Acid	26% Gelled Acid	2060	2060	8.2	11837.11	8.2	12098.05
8	Spacer	Linear Gel 30#	500	500	8.3	10639.63	10.8	10883.47
9	Self-degradable solid Diverter	Linear Gel 30# + NWB	100	100	10.8	10726.72	10.8	10854.75
10	Spacer	Linear Gel 30#	500	500	9.8	10017.61	10.8	10605.20
11	Pad	30# x-linked Gel	2300	2325	15.8	11091.47	18.2	11885.01
12	Acid	26% Emulsified Acid	4580	4580	17.3	11706.82	18.1	11933.75
13	Pad	30# x-linked Gel	1000	1011	19.5	11908.38	21.3	12234.69
14	Acid	26% Gelled Acid	2060	2060	19.5	11854.61	20.6	12160.58
15	Far field Diverter	RPM	3000	3000	19.0	11693.39	19.7	11866.86
16	Pad	30# x-linked Gel	2300	2318	19.6	11879.30	19.9	11960.95
17	Acid	26% Emulsified Acid	4580	4580	20.8	11900.37	25.6	12196.05
18	Pad	30# x-linked Gel	1000	1008	24.9	12221.45	24.9	12282.48
19	Acid	26% Gelled Acid	2060	2060	17.9	10964.40	24.9	12452.37
20	Spacer	Linear Gel 30#	500	500	9.7	9065.11	11.8	9312.49
21	Self-degradable solid Diverter	Linear Gel 30# + NWB	150	150	10.8	9175.15	10.8	9199.69
22	Spacer	Linear Gel 30#	500	500	10.0	9132.42	10.8	9238.80
23	Pad	30# x-linked Gel	2300	2320	18.2	10726.07	21.8	11714.22
24	Acid	26% Emulsified Acid	4580	4580	21.8	11964.66	21.8	12122.43
25	Pad	30# x-linked Gel	1000	1009	23.3	11877.01	25.1	12230.39
26	Acid	26% Gelled Acid	2060	2060	24.8	11947.40	24.9	12153.63
27	Far field Diverter	RPM	3500	3500	24.1	11997.48	24.9	12446.51
28	Pad	30# x-linked Gel	2300	2314	22.5	12009.92	23.3	12227.14
29	Acid	26% Emulsified Acid	4580	4580	22.9	12370.67	22.9	12500.19
30	Pad	30# x-linked Gel	1000	1006	22.9	12465.37	23.1	12511.87
31	Acid	26% Gelled Acid	2060	2060	22.9	12382.29	23.0	12490.55
32	Acid	Tank Bottoms	3780	3780	22.9	12345.75	22.9	12387.11
33	Flush	Treated Water	12516	12516	22.3	12516.78	22.9	12782.23
34	Overflush	Treated Water	3755	3755	21.2	12299.63	21.2	12320.51
Total			82501	82659				

Table 3—Stimulation fluid volume for stage-1 (lower stage / stage-1)

Fluid #	Total Volume	Volume (gallon)	Volume (bbl.)
1	Crosslinked gel 30#	13,360	318
2	HCl 26%	12,688	302
3	Emulsified Acid	22,190	552
4	Diverter	6,632	158
5	Solid diverter	500 (lb)	-
6	Treated DPW Water	18,466	440

The 1st stage was temporarily isolated using a frac ball and simultaneously activated the 2nd stage frac port in the lower stage. Similar steps were followed in the 2nd stage for acid stimulation using DPW. Table 4 and 5 show the 2nd stage pumping schedule, fluids and pressure/rate summary.

Table 4—Pumping schedule for stage-2 (Well A - lower stage / stage-1)

Stage Number	Description	Fluid System	Clean Volume gal	Slurry Volume gal	Avg Slurry Rate bpm	Avg Treating Pressure psi	Max Slurry Rate bpm	Max Treat Pr psi
1	Spacer	Treated Water	500	500	6.5	5012.94	10.0	5874.28
2	Acid	26% Gelled Acid	2000	2000	14.0	7274.07	15.0	7954.39
3	Spacer	Treated Water	1500	1500	16.2	8361.46	18.8	9217.57
4	Pad	30# x-linked Gel	2300	2333	19.7	10087.22	20.0	10480.84
5	Acid	26% Emulsified Acid	4580	4580	20.0	10987.13	20.0	11545.88
6	Pad	30# x-linked Gel	1000	1014	20.0	11667.88	20.0	11755.79
7	Acid	26% Gelled Acid	2060	2060	14.5	10424.24	20.0	11916.55
8	Spacer	Linear Gel 30#	500	500	9.2	8487.29	10.8	8744.23
9	Self-degradable solid Diverter	Linear Gel 30# + NWB	100	100	10.8	8732.15	10.8	8739.82
10	Spacer	Linear Gel 30#	500	500	9.7	8578.50	10.8	8718.37
11	Pad	30# x-linked Gel	2300	2325	18.5	10308.04	24.3	11820.88
12	Acid	26% Emulsified Acid	4580	4580	22.7	12300.05	23.6	12706.87
13	Pad	30# x-linked Gel	1000	1011	21.7	12604.43	21.7	12655.56
14	Acid	26% Gelled Acid	2060	2060	21.9	12484.42	25.4	12647.72
15	Far field Diverter	RPM	3000	3000	19.4	11644.97	25.2	12774.26
16	Pad	30# x-linked Gel	2300	2318	20.0	12171.36	20.0	12221.44
17	Acid	26% Emulsified Acid	4580	4580	20.0	11953.61	20.0	12187.70
18	Pad	30# x-linked Gel	1000	1008	20.0	12022.29	20.0	12046.08
19	Acid	26% Gelled Acid	2060	2060	15.6	11201.95	20.2	12018.72
20	Spacer	Linear Gel 30#	500	500	14.5	10566.21	15.8	10733.35
21	Self-degradable solid Diverter	Linear Gel 30# + NWB	150	150	10.8	10063.47	10.8	10087.49
22	Spacer	Linear Gel 30#	500	500	9.7	9919.57	10.8	10065.16
23	Pad	30# x-linked Gel	2300	2320	16.6	10447.17	20.4	11106.09
24	Acid	26% Emulsified Acid	4580	4580	21.3	11727.24	23.5	12616.75
25	Pad	30# x-linked Gel	1000	1009	23.5	12562.72	23.6	12619.77
26	Acid	26% Gelled Acid	2060	2060	23.5	12192.97	23.6	12440.68
27	Far field Diverter	RPM	3500	3500	23.5	11887.33	23.6	12143.41
28	Pad	30# x-linked Gel	2300	2314	23.5	12225.84	23.6	12275.82
29	Acid	26% Emulsified Acid	4580	4580	23.5	12410.70	23.6	12532.32
30	Pad	30# x-linked Gel	1000	1006	22.5	12276.91	23.6	12545.56
31	Acid	26% Gelled Acid	2060	2060	23.5	12471.82	23.6	12534.37
32	Acid	Tank Bottoms	3780	3780	23.5	12252.53	23.6	12509.88
33	Flush	Treated Water	12368	12368	22.8	12180.33	23.6	12709.30
34	Overflush	Treated Water	3710	3710	20.4	12373.55	20.4	12452.30

Table 5—Fluids, Rate and Pressure Summary for stage-2 (Well A -upper stage / stage-2)

	Injection Test	Main FRAC
Fluid Summary		
Crosslinked gel 40#, bbl	-	379
Linear gel 40#, bbl		61
HCl 26%, bbl	-	288
Emulsified Acid, bbl	-	537
Diverter, bbl	-	152
Treated DPW, bbl	56	439
Pressure and rate summary		
Well Initial shut-in press, psi	4,911	4,291
Max Pumping rate, bpm	18	25.4
Avg pumping rate, bpm	10.4	19.6
Max STP, psi	11,967	12,774
Avg STP, psi	7,674	11,190
Max. BHP, psi	16,531	16,892
BH ISIP, psi	16,005	15,648
Surface ISIP, psi	10,003	9,575
5 min ISIP, psi	6,711	9,277
10 min ISIP, psi	6,483	9,052
15 min ISIP, psi	6,331	8,883

Fluid #	Total Volume	Volume (gallon)	Volume (bbl)
1	Crosslinked gel 40#	15,919	379
2	Linear gel 40#	2,580	61
3	HCl 26%	12,094	288
4	Emulsified Acid	22,544	379
5	Diverter	6,377	152
6	Solid particulate diverter	500	-
7	Treated DPW Water	18,435	439

Evaporation Pond Water

The water quantity and quality of the pond, available at a plant, has been evaluated for re-use opportunity, especially for frac operations, with favorable geochemical composition. An average of 17,000 bbl. of water per day is being discharged to evaporation ponds in the gas plant. The source of this water comes from different feeds; yet, the boiler blowdown and co-generation plant water consistently makes up about 60%-70% of the total gas plant evaporation pond water. The pond water was used to acid stimulate two stages in the upper and lower carbonate reservoirs of Well B. 100% of the ground water requirement was preserved by completely replacing it with gas plant evaporation pond water during the acid stimulation treatments. Zirconate crosslinked gel is formulated using the industrial water from the evaporation ponds in gas plant that addressed the desired stimulation fluids characteristics for enhanced reservoir stimulation. Several rheology and core flood tests were performed to test the potential formation damage as well as the stability of such fluid under high temperature environment.

Well B. Well B was drilled as a vertical gas producer across both two carbonate formations and completed with a monobore cemented liner along with permanent packer and polished bore receptacle seals. The well was perforated and stimulated in two stages in the lower and upper carbonate intervals. Well B was selected as the candidate for pond water-based stimulation fluids for the following reasons:

- Satisfactory and comparable geo-chem analysis to the ground water
- Pond water-based PAD fluid developed satisfactory rheology at reservoir conditions using multiple shear rates.
- Pond water-based gel exhibited high retained permeability with synthetic carbonate core.
- Pond water-based gel was formulated with an appropriate scale inhibitor that exhibited no precipitation of scale.

Fracturing Fluids Description. Acid fracturing treatment in carbonate formations included alternating crosslinked gel (PAD), regular HCl (28%), emulsified acid (28%) and diverter. The PAD fluid used to create the fracture profile and fingering effect is zirconate crosslinked gel. Acid volumes during acid fracture treatment ranged from 400 to700 gal per feet of net pay.

The laboratory tests were performed focusing on:

- Stability and rheology profile of zirconate crosslinked gel at higher than 300°F (Table-6 and Fig-6).
- Regained permeability using the broken gel system.
- Stability at room temperature and downhole temperatures for 28% HCl regular acid, emulsified acid and visco-elastic surfactant based self-diverted acid.
- Rheological performance of spent diverter at surface and downhole conditions.

Laboratory Results.
Benchtop QAQC:

Table 6—Benchtop QAQC of Gel at Room Temperature

Test Item	40# Linear gel	45# Linear gel
Linear Gel 3 minutes Viscosity@511 s^{-1}	40	45
Linear Gel 10 minutes Viscosity@511 s^{-1}	41	46
Linear Gel Temperature	75°F	76°F
Linear Gel pH	7.59	7.56
pH after addition of buffer BF-9L	10.22	10.23
pH after addition of delayed crosslinker	9.93	9.97

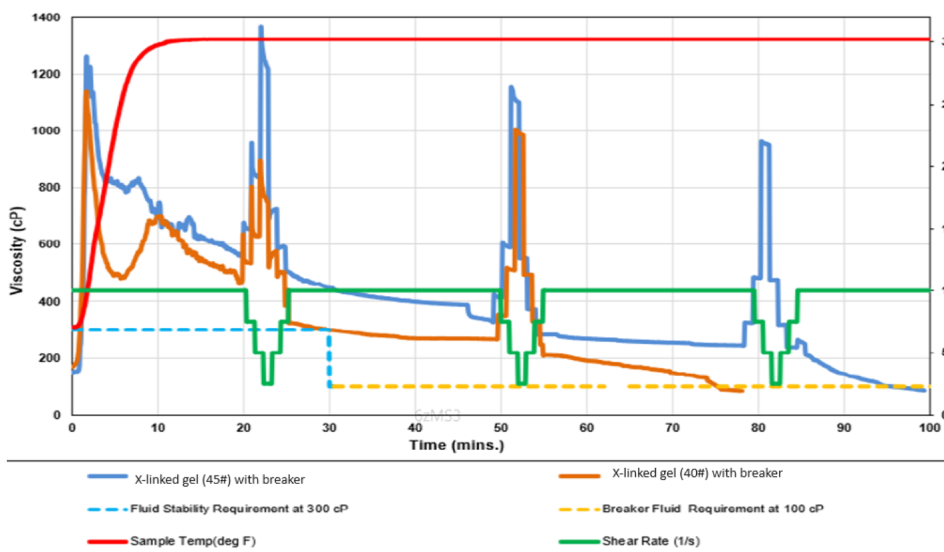

Figure 6—Rheology of crosslinked at 302 deg F

Stimulation Design and Execution. Design: The design includes 28% hydrochloric acid (HCl) and emulsified acid as main reactive treatment fluid using an average combined volume of 600-gallon acid per ft. of perforation. The zirconate based crosslinked gel as Pad was optimized with pond water (PW). The Acid to PAD ratio was selected to be 5:4 based on reservoir petrophysical and lithology properties. A compatible scale inhibitor between the formation brine and pond water was identified and included in the gel formulation. To minimize fluid leak off to high permeable zones and improve diversion effects, visco-elastic surfactant-based diverter was included in the stimulation design.

Execution (Lower Stage / stage-1):

Initial WH pressure was 1,400 psi. Injection test (Fig-7) started by pumping treated water at 1.0 bpm and then rate was increased to maximum of 15 bpm. Total volume of treated PW pumped was 100 bbl. The maximum STP was 9,425 psi and maximum BHTP was 14,900 psi. The bottomhole Closure Pressure was determined to be 7,909 psi (Fig-8).

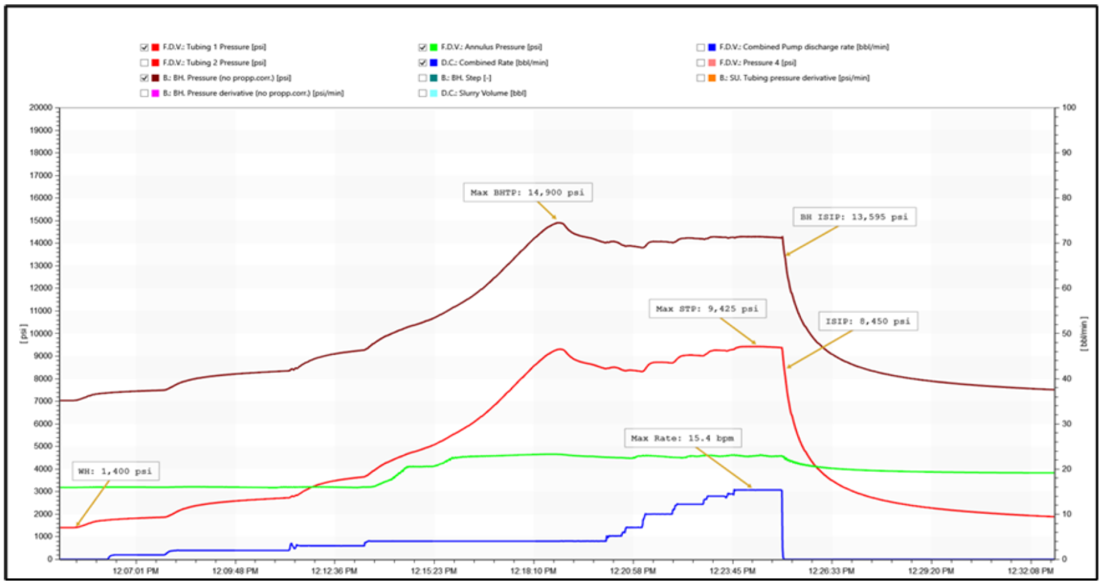

Figure 7—Injection Test (lower stage / stage-1)

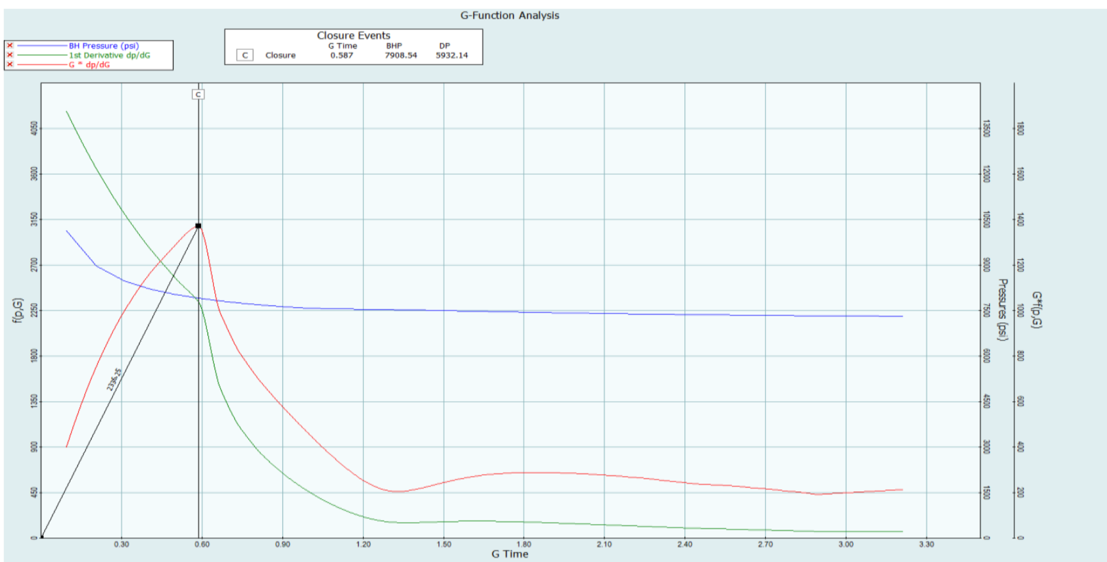

Figure 8—Closure analysis (lower stage / stage-1)

The PW trial test was performed for Well-B lower stage starting at a rate of 10.0 bpm with stable surface and bottomhole treatment pressures. After initial acid volumes reached formation, pumping rate was increased to max 43 bpm. Total fluid pumped was 1,536 bbl. Acid was flushed and over flushed with 321 bbl of treated PW. Maximum STP was 11,730 psi while the maximum BHTP was 14,665 psi. Table 7 represents the final bottomhole pumping schedule, showing the alternating stages of Pad, acid and diverter pills.

Table 7—Pumping Schedule (Lower Stage / Stage-1)

As Measured Pump Schedule							
Step #	Step Name	Fluid Name	Slurry Volume	Slurry Rate	Average Pressure	Pump Time	Fluid Volume
			(bbl)	(bbl/min)	(psi)	(min)	(gal)
1	Establish Injection	Treated Pond Water	22.0	8.7	5,220.0	2.5	924
2	Spearhead Acid	28% HCL (275F)	48.0	10.1	8,372.0	4.8	2016
3	Acid Displacement	Treated Pond Water	100.0	15.5	8,827.0	6.5	4200
4	PAD-1	Crosslinked gel 40#	83.0	18.8	9,048.0	4.4	3486
5	Emulsified Acid-1	Emulsified Acid	50.0	26.7	10,025.0	1.9	2100
6	Straight acid-1	28% HCL (275F)	61.0	30.0	9,879.0	2.0	2562
7	Diversion-1	Diverter	42.0	30.0	10,057.0	1.4	1764
8	Spacer Water-1	Spacer Water	25.0	30.0	10,045.0	0.8	1050
9	PAD-2	Crosslinked gel 40#	86.0	34.3	10,766.0	2.5	3612
10	Emulsified Acid-2	Emulsified Acid	50.0	39.0	10,076.0	1.3	2100
11	Straight acid-2	28% HCL (275F)	63.0	42.2	10,317.0	1.5	2646
12	Diversion-2	Diverter	42.0	43.0	10,391.0	1.0	1764
13	Spacer Water-2	Spacer Water	25.0	43.0	10,637.0	0.6	1050
14	PAD-3	Crosslinked gel 40#	86.0	41.1	11,232.0	2.1	3612
15	Emulsified Acid-3	Emulsified Acid	50.0	41.7	10,181.0	1.2	2100
16	Straight acid-3	28% HCL (275F)	63.0	43.0	8,816.0	1.5	2646
17	Diversion-3	Diverter	42.0	43.0	8,013.0	1.0	1764
18	Spacer Water-3	Spacer Water	25.0	43.0	8,086.0	0.6	1050
19	PAD-4	Crosslinked gel 40#	88.0	43.0	8,368.0	2.0	3696
20	Emulsified Acid-4	Emulsified Acid	50.0	43.0	7,226.0	1.2	2100
21	Straight acid-4	28% HCL (275F)	65.0	43.0	6,334.0	1.5	2730
22	Sweep	Crosslinked gel 40#	49.0	43.0	6,328.0	1.1	2058
23	75% Overflush	Treated Pond Water	107.0	43.0	7,143.0	2.5	4494
24	Flush	Treated Pond Water	214.0	24.7	4,424.0	8.7	8988

The actual pumped stimulation fluids volume (Table 8) was very close to the treatment design. Table-9 shows the summary of fluids, rate and pressure values of stage-1 stimulation treatment.

Table 8—Stimulation Fluids Volume (Lower Stage / stage-1)

Fluid #	Total Volume	Volume (gallon)	Volume (bbl)
1	Crosslinked gel 40#	16,464	392
2	HCl 28%	12,600	300
3	Emulsified Acid	8,400	200
4	Diverter	5,292	126
5	Spacer Water	3,150	75
6	Treated Pond Water	18,606	443

Table 9—Treatment Parameters Summary (Well B lower stage / stage-1)

	Injection Test	Main FRAC
Fluid Summary		
Crosslinked gel 40#, bbl	-	392
HCl 28%, bbl	-	300
Emulsified Acid, bbl	-	200
Diverter, bbl	-	126
Spacer Water, bbl	-	75
Treated Pond Water, bbl	100	443
Pressure and rate summary		
Well Initial shut-in press, psi	1,400	1,170
Max Pumping rate, bpm	15.4	43.0
Avg pumping rate, bpm	5.3	29.0
Max STP, psi	9,425	11,730
Avg STP, psi	5,700	8,502
Max. BHP, psi	14,900	14,665
Avg BHP, psi	11,220	11,992
BH ISIP, psi	13,595	7,113
Surface ISIP, psi	8,450	1,461
5 min ISIP, psi	2,155	1,412
10 min ISIP, psi	1,753	1,380
15 min ISIP, psi	1,672	1,360
Shut in, psi	-	-
BH Closure Pressure, BHP	7,908	-
FE %	22.7	-
Closure gradient, psi/ft	0.60	-

The lower stage was temporarily isolated and total 35 ft interval was perforated in the upper stage. Similar steps were followed in the upper stage for acid stimulation using PW. Table 10 and 11 show the upper stage pumping schedule, fluids and pressure/rate summary.

Table 10—Pumping Schedule (upper stage / stage-2)

As Measured Pump Schedule							
Step #	Step Name	Fluid Name	Slurry Volume	Slurry Rate	Average Pressure	Pump Time	Fluid Volume
			(bbl)	(bbl/min)	(psi)	(min)	(gal)
1	Establish Injection	Treated Pond Water	15.0	1.8	5,737.0	8.3	630
2	Spearhead Acid	28% HCL (275F)	48.0	1.8	10,680.0	26.7	2016
3	Acid Displacement	Treated Pond Water	226.0	2.0	10,713.0	113.0	9492
4	PAD-1	Linear Gel 40#	83.0	23.0	11,505.0	3.6	3486
5	Emulsified Acid-1	Emulsified Acid	50.0	22.0	11,800.0	2.3	2100
6	Straight acid-1	28% HCL (275F)	61.0	19.2	11,802.0	3.2	2562
7	Diversion-1	Diverter	42.0	18.2	11,812.0	2.3	1764
8	Spacer Water-1	Spacer Water	25.0	11.2	11,113.0	2.2	1050
9	PAD-2	Crosslinked gel 40#	86.0	9.6	10,723.0	9.0	3612
10	Emulsified Acid-2	Emulsified Acid	50.0	17.7	10,861.0	2.8	2100
11	Straight acid-2	28% HCL (275F)	63.0	26.1	10,823.0	2.4	2646
12	Diversion-2	Diverter	42.0	28.1	10,968.0	1.5	1764
13	Spacer Water-2	Spacer Water	25.0	28.1	11,156.0	0.9	1050
14	PAD-3	Crosslinked gel 40#	86.0	28.1	11,282.0	3.1	3612
15	Emulsified Acid-3	Emulsified Acid	50.0	28.3	11,142.0	1.8	2100
16	Straight acid-3	28% HCL (275F)	63.0	30.4	11,019.0	2.1	2646
17	Diversion-3	Diverter	42.0	31.2	10,895.0	1.3	1764
18	Spacer Water-3	Spacer Water	25.0	31.2	10,964.0	0.8	1050
19	PAD-4	Crosslinked gel 40#	88.0	31.2	11,225.0	2.8	3696
20	Emulsified Acid-4	Emulsified Acid	50.0	32.6	10,554.0	1.5	2100
21	Straight acid-4	28% HCL (275F)	65.0	19.5	8,236.0	3.3	2730
22	Tank Bottoms	28% HCL (275F)	145.0	10.0	5,842.0	14.5	6090
23	Sweep	Crosslinked gel 40#	50.0	27.7	8,218.0	1.8	2100
24	75% Overflush	Treated Pond Water	107.0	42.1	11,310.0	2.5	4494
25	Flush	Treated Pond Water	212.0	38.5	11,446.0	5.5	8904

Table 11—Fluids, Rate and Pressure Summary (upper stage / stage-2)

	Injection Test	Main FRAC
Fluid Summary		
Crosslinked gel 40#, bbl	-	310
Linear gel 40#, bbl		83
HCl 28%, bbl	-	445
Emulsified Acid, bbl	-	200
Diverter, bbl	-	126
Spacer Water, bbl	-	75
Treated Pond Water, bbl	101	560
Pressure and rate summary		
Well Initial shut-in press, psi	1,835	2,090
Max Pumping rate, bpm	3.0	43.0
Avg pumping rate, bpm	2.2	8.4
Max STP, psi	10,886	11,880
Avg STP, psi	9,597	10,338
Max. BHP, psi	16,330	16,500
Avg BHP, psi	15,053	15,454
BH ISIP, psi	16,196	11,950
Surface ISIP, psi	10,805	6,870
5 min ISIP, psi	6,884	5,140
10 min ISIP, psi	5,770	4,730
15 min ISIP, psi	5,209	4,350
Shut in, psi	3,337	3,735
BH Closure Pressure, BHP	12,205	
FE %	11.9	
Closure gradient, psi/ft	0.9	

Fluid #	Total Volume	Volume (gallon)	Volume (bbl)
1	Crosslinked gel 40#	13,020	310
2	Linear gel 40#	3,486	83
3	HCl 28%	18,690	445
4	Emulsified Acid	8,400	200
5	Diverter	5,292	126
6	Spacer Water	3,150	75
7	Treated Pond Water	23,520	560

Nano-Filtered (NF) Seawater Based Stimulation Fluids

Several nano-filtered seawater based fracturing fluids have been successfully trial tested in the carbonate formation after confirming adequate stable rheology and formation compatibility at the reservoir conditions. Recently, inhouse nano-filtered seawater based crosslinked gel has been successfully formulated to provide required stimulation fluids characteristics. The acid blends formulated with NF seawater with standard corrosion inhibition package for high temperature didn't show any signs of incompatibility such as fluid phase separation, flocculation, or precipitation.

Well C. Well C was drilled as a vertical gas producer across both carbonate formations and completed with a monobore cemented liner along with permanent packer and polished bore receptacle seals. The well was perforated and stimulated in two stages in the lower and upper carbonate intervals. Well C was selected as the candidate for NF seawater-based stimulation fluids for the following reasons:

- Significant reduction in the Sulfate content and favorable geo-chem results after nano-filtration.
- NF PAD fluid developed satisfactory rheology at reservoir conditions using multiple shear rates.
- NF formulated with an appropriate scale inhibitor that exhibited no precipitation of scale.

Fracturing Fluids Description. Acid fracturing treatment in carbonate formations included alternating crosslinked gel (PAD), Regular HCl (28%), emulsified acid (28%) and diverter. The PAD fluid used to create the fracture profile and fingering effect is zirconate crosslinked gel. Current acid volumes during acid fracture treatment range from 400 to 700 gal per feet of net pay.

The laboratory tests were performed focusing on:

- Stability and viscosity breaking of zirconate crosslinked gel at higher than 300°F.
- Stability of 28% HCl straight acid, emulsified acid and visco-elastic surfactant based self-diverting acid.
- Rheological performance of spent diverter at surface and BH conditions.

Laboratory Results.

Nano Filtered Seawater GEOC:

GEOC analysis was performed which exhibited most of the cationic values are within the acceptable (Table-12). The sulfate content is very low (16 mg/l) which represents the effectiveness of the nanofiltration.

Table 12—GEOC analysis

Parameters	Limit	Nano-filtered Seawater	Unit
Fluid Test Temperature (degF)	35-100	81.4	deg F
Color (unfiltered sample)	Clear	Clear	
pH	6.0-8.0	7.64	pH
Specific Gravity	0.99-1.01	1.03	g/cm^3
Chloride, Cl^-	<30,000	32390	mg/l
Bicarbonate, HCO_3^{-2}	<500	1446	mg/l
Iron, Total	<10	0.19	mg/l
Sulfate	<500	16	mg/l
Total Harness	<500	694	mg/l
Calcium Hardness	<500	403	mg/l
Magnesium Hardness	<500	291	mg/l
TDS	-	44	g/l

HPHT rheometer test conducted on NF seawater-based cross-linked gel showed stabilized viscosity profile and favorable breaking behavior at temperature higher than 300°F and under multiple shear rates. Intensive lab tests were conducted to test the stability of NF SW with frac fluids at high temperature environment. The NF SW is compatible with the frac fluids with no detected separation or precipitation in the fluids. Also, the drop test/ snake test showed no dispersing out in the emulsified acid.

Stimulation Design and Execution. Design: The design includes 28% hydrochloric acid (HCl) and emulsified acid as main reactive treatment fluid using an average combined volume of 600-gallon acid per ft. of perforation. The zirconate based crosslinked gel as Pad was optimized with Nano-filtered seawater. The Acid to PAD ratio was selected to be 3:2 based on reservoir petrophysical and lithology properties. A compatible scale inhibitor between the formation brine and NF seawater was identified and included in the gel formulation. To minimize fluid leak off to high permeable zones and improve diversion effects visco-elastic surfactant-based diverter was included in the stimulation design.

Execution (Lower formation):

For water hauling, an adapter flange was installed to the injection line. A 2 in. treatment line was rigged up from the injection line to outside of the fence. This is in order to have an easy water hauling with no interruption going inside the injector well location Fig-9.

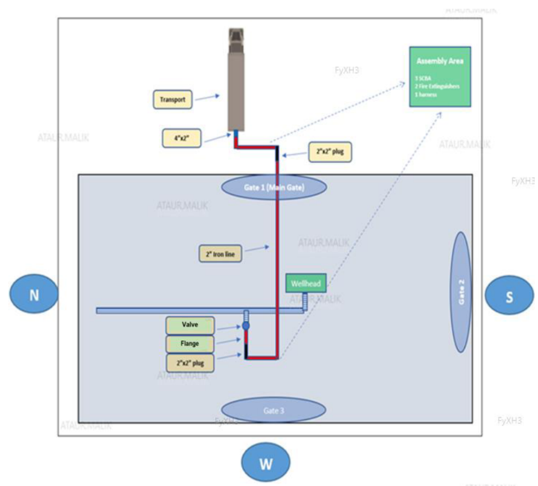

Figure 9—NF SW Hauling Rig Up

Initial WH pressure was 4,500 psi. Injection test (Fig-10) started by pumping treated water at 1.0 bpm and then rate was increased to max 15.0 bpm. Total volume of NF seawater pumped was 320 bbl. The maximum

STP was 10,813 psi and maximum BHTP was 15,998 psi. The NF seawater trial test was performed in Well-C (Fig-11) lower formation starting at a rate of 8.0 bpm with stable surface and bottomhole treatment pressures. After initial acid volumes reached formation, pumping rate was increased to max 42 bpm. Total fluid pumped was 1,609 bbl. Acid was flushed and over flushed with 321 bbl of treated NF seawater.

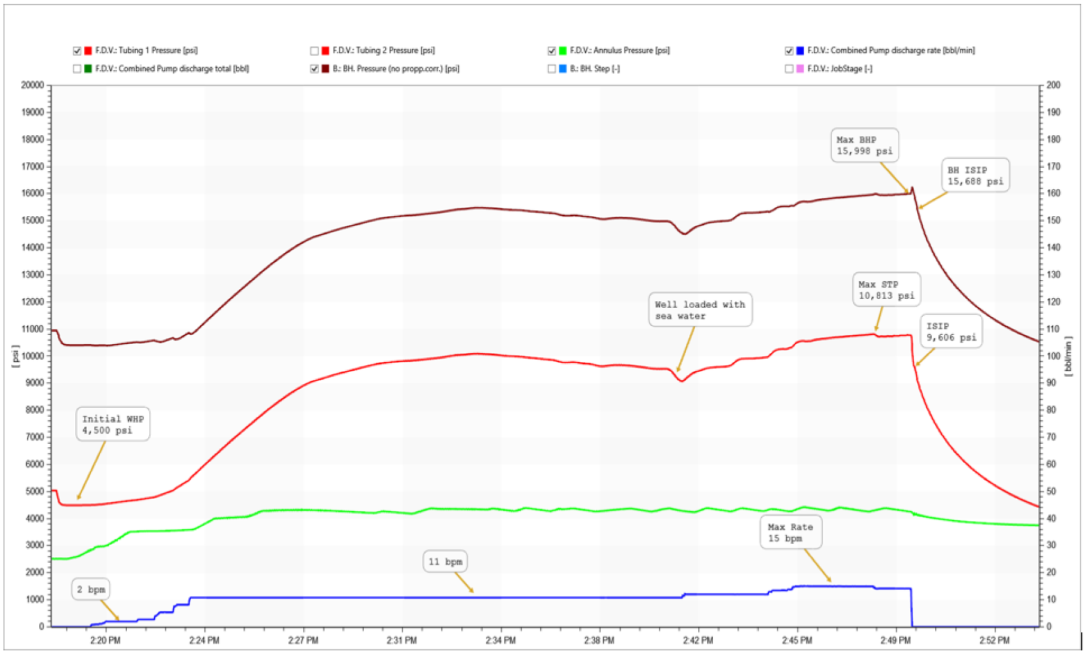

Figure 10—NF seawater injection test in Well C (lower stage / stage-1)

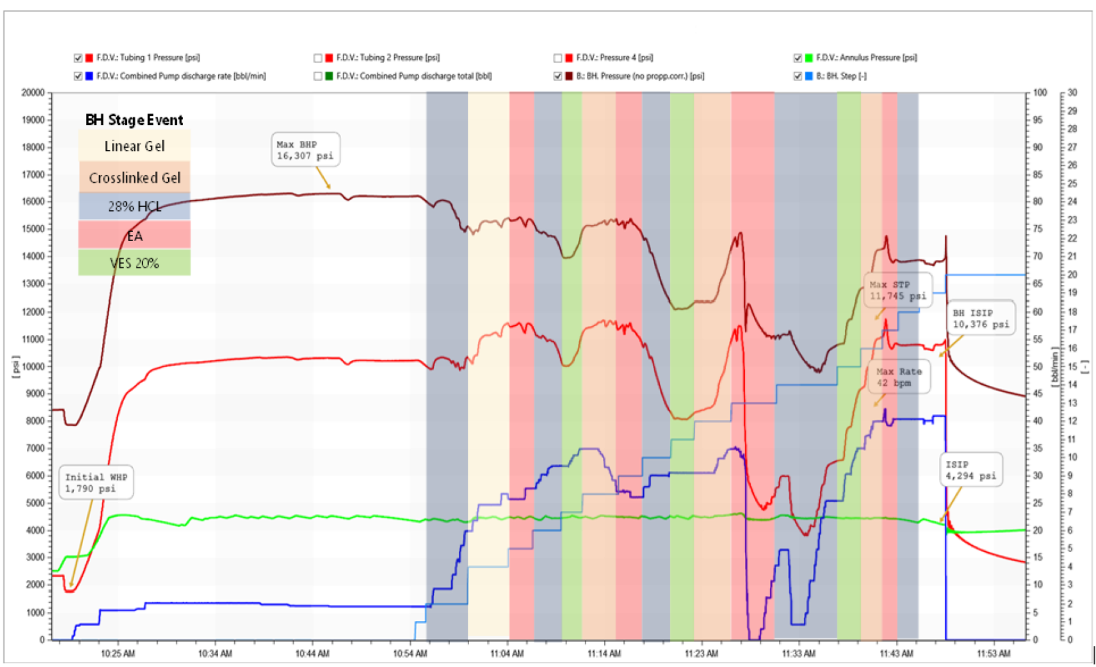

Figure 11—NF seawater-based acid stimulation in Well C (lower stage / stage-1)

Table-13 represents the final bottomhole pumping schedule showing the alternating stages of Pad, acid and diverter.

Table 13—Pumping Schedule (lower stage / stage-1)

As Measured Pump Schedule						
Step #	Step Name	Fluid Name	Slurry Volume	Slurry Rate	Pump Time	Average Pressure
			(bbl)	(bbl/min)	(min)	(psi)
1	Establish Injection	Nano-filtered SW	8.0	2.6	3.1	2,406
2	Spearhead Acid	Acid 28%	47.6	6.1	7.8	8,637
3	Acid Displacement	Nano-filtered SW	0.0	6.9	0.0	10,090
4	PAD-1	NFSW Gel-40 (LINEAR)	95.2	6.7	14.2	10,252
5	Delayed Acid-1	EA	64.3	6.2	10.4	10,237
6	Straight Acid-1	Acid 28%	85.7	13.0	6.6	10,238
7	Diversion-1	Diverter	71.4	25.0	2.9	11,284
8	PAD-2	NFSW X-linked Gel-40	119.0	28.0	4.3	11,293
9	Delayed Acid-2	EA	64.3	32.0	2.0	10,309
10	Straight Acid-2	Acid 28%	85.7	35.0	2.4	11,133
11	Diversion-2	Diverter	71.4	31.0	2.3	11,581
12	PAD-3	NFSW X-linked Gel-40	119.0	28.0	4.3	10,712
13	Delayed Acid-3	EA	64.3	30.0	2.1	8,301
14	Straight Acid-3	Acid 28%	85.7	31.0	2.8	8,318
15	Diversion-3	Diverter	71.4	33.0	2.2	9,743
16	PAD-4	NFSW X-linked Gel-40	84.7	14.0	6.1	6,162
17	Delayed Acid-4	EA	64.3	18.0	3.6	5,225
18	Straight Acid-4	Acid 28%	85.7	32.0	2.7	8,076
19	75% Overflush	Nano-filtered SW	107.0	39.0	2.7	10,736
20	Flush	Nano-filtered SW	214.3	40.0	5.4	10,761

The actual pumped stimulation fluids volume was very close to the treatment design. Table-14 shows Nano-Filtered Seawater (NFSW) based totalizers.

Table 14—Stimulation Fluids Volume (Lower stage / stage-1)

Fluid #	Total Volume	Volume (gallon)	Volume (bbl)
1	NFSW Crosslinked gel 40#	13,557	323
2	HCl 28%	16,400	390
3	Emulsified Acid	10,800	257
4	Diverter	9,000	214
5	NFSW Liner gel 40#	4,000	95
6	Nano-filtered SW	13,830	329

The lower formation was temporarily isolated and total 68 ft interval was perforated in the upper formation. Similar steps were followed in the upper formation for acid stimulation using NFSW based stimulation fluids. Table 15 and 16 show the upper formation pumping schedule, fluids and pressure/rate summary.

Table 15—Pumping Schedule (upper stage / stage-2)

As Measured Pump Schedule						
Step #	Step Name	Fluid Name	Slurry Volume	Slurry Rate	Pump Time	Average Pressure
			(bbl)	(bbl/min)	(min)	(psi)
1	Establish Injection	Nano-filtered SW	5.7	2.5	2.3	3,890
2	Spearhead Acid	Acid 28%	75.0	6.3	11.9	7,188
3	Acid Displacement	Nano-filtered SW	159.8	2.7	59.2	10,271
4	PAD-1	NFSW Gel-40 (LINEAR)	47.6	12.0	4.0	10,429
5	Delayed Acid-1	EA	31.0	17.0	1.8	10,820
6	Straight Acid-1	Acid 28%	79.0	20.0	4.0	11,322
7	Diversion-1	Diverter	55.0	20.0	2.8	11,441
8	PAD-2	NFSW X-linked Gel-40	81.0	20.0	4.1	11,434
9	Delayed Acid-2	EA	42.9	20.0	2.1	11,377
10	Straight Acid-2	Acid 28%	79.0	20.0	4.0	11,425
11	Diversion-2	Diverter	55.0	21.0	2.6	11,298
12	PAD-3	NFSW X-linked Gel-40	81.0	21.0	3.9	11,377
13	Delayed Acid-3	EA	45.2	22.0	2.1	11,263
14	Straight Acid-3	Acid 28%	79.0	22.0	3.6	11,385
15	Diversion-3	Diverter	55.0	21.0	2.6	11,323
16	PAD-4	NFSW X-linked Gel-40	119.0	21.0	5.7	11,273
17	Delayed Acid-4	EA	71.4	22.0	3.2	11,072
18	Straight Acid-4	Acid 28%	92.6	20.0	4.6	11,478
19	TANK BOTTOM	Acid 28%	58.5	12.0	4.9	10,285
20	75% Overflush	Nano-filtered SW	150.0	18.0	8.3	11,166
21	Flush	Nano-filtered SW	214.3	21.0	10.2	11,236

Table 16—Fluids, Rate and Pressure Summary (upper stage / stage-2)

	Injection Test	Main FRAC
Fluid Summary		
NFSW crosslinked gel 40#, bbl	-	281
NFSW linear gel 40#, bbl		48
HCl 28%, bbl	-	463
Emulsified Acid, bbl	-	190
Diverter, bbl	-	165
NFSW , bbl	250	530
Pressure and rate summary		
Well Initial shut-in press, psi	2,800	3,650
Max Pumping rate, bpm	10.5	23.0
Avg pumping rate, bpm	6.3	11.3
Max STP, psi	10,712	11,615
Avg STP, psi	9,523	10,379
Max. BHP, psi	16,361	16,407
Avg BHP, psi	15,016	15,264
BH ISIP, psi	16,189	15,264
Surface ISIP, psi	10,318	9,393
5 min ISIP, psi	9,423	8,978
10 min ISIP, psi	8,845	8,732
15 min ISIP, psi	5,209	4,570

Fluid #	Total Volume	Volume (gallon)	Volume (bbl)
1	NFSW crosslinked gel 40#	11,800	281
2	NFSW linear gel 40#	2,000	48
3	HCl 28%	19,457	463
4	Emulsified Acid	8,000	190
5	Diverter	6,930	165
6	NFSW	22,254	530

Post Job Flow Back Results

The wells' performance during flowback operations indicates successful outcome following stimulation using Pond water, Distilled Produced Water, and Nano-filtered Seawater. The candidate wells showed no sign of formation damage and productivity index (PI) comparable to nearby wells. Flowback was successful, indicating effective stimulation without immediate issues. Post-clean-up, each well was tested at three stable rates, showing no formation damage, which is crucial for maintaining productivity. The PI of 2-2.3 SCFD/ psi^2 aligns with offset wells, suggesting effective stimulation.

Conclusions

Distilled Produced Water (DPW) from GOSP was successfully pumped for two stages in Well A as part of the reduced ground water initiative under the sustainability approach. All acid blends (emulsified and HCl) as well as PAD (Zirconate Crosslinked Gel) and linear gel were mixed using DPW during acid fracturing treatments in stage 1 and 2. The treatments have been pumped without any changes applied to the initial design. All the fluids had been tested for rheology, compatibility, dissolution etc., prior to the job execution, and the fluids formulation had been optimized accordingly. both surface and BH pressures showed positive effect of the treatment and good indication of the chemical reaction between acid and reservoir. Total ground water saved during stage 1 and 2 are approximately 520 bbl and 475 bbl, respectively.

In line with sustainability approach to protect environment and conserve fresh water all stakeholders collaborated to field implement fracturing fluids formulated using discharged water to the evaporation ponds . Two stages acid fracture treatments have been successfully executed in the upper and lower formations of Well B using plug and perf techique and PW based fracturing fluids. Approximately, 2,100 bbl of ground water was preserved by completely replacing ground water with evaporation PW during the acid stimuation treatments in two stages. The fluid formulation using PW is developed with desired characteristics for enhanced reservoir stimulation. Several rheology and core flooding tests were performed to test the potential formation damage as well as the stability of such fluid at high temperature environment. The pond water successfully sustained viscosity values at required level.

In line with sustainability approach to further protect environment and conserve fresh water all stakeholders collaborated to field implement fracturing fluids formulated with nano-filtered (NF) seawater. Two stages acid fracture treatments have been successfully executed in the carbonate reservoirs of Well C using P&P techique and NF seawater based fracturing fluids. Approximately, 2,000 bbl of ground water was preserved by completely replacing ground water with NF seawater during the acid stimuation treatments in stage 1 and stage 2. The NF seawater was collected from a nearby water injecton line. NF seawater based stimulation fluid is developed to address the water conservation initiative and enhance reservoir stimulation. Intensive laboratory tests were conducted to test the potential formation damage as well as the stability of such fluid at high temperature environment. The NF seawater based gel satisfactorily sustained viscosity values at required level.

Recommendations

In all fresh water substitutes, the effectiveness of the scale inhibitors needs to be assessed in the long run by performing well testing, production logging etc. The PW quality may vary with various seasons. Therefore, the geo-chemical analysis will need to be performed intensively on the PW samples and if needed, the PW should be chemically treated or nano-filtered prior to its application.

The application of fresh water substitutes can be extended to proppant fracturing treatment in the clastic reservoirs with additional laboratory tests.

For large scale implementation, fresh water substitutes may require investment in the infrastructure (effective loading points, central storage points, mobile filtration and water treatment units etc.) to load and transfer them to the well locations in an operationally efficient manner.

Similar approach can be extended to other GOSPs and evaporation ponds of different gas plants, water injector wells for NF seawater to secure more volume of fresh water substitutes and additional flexibilities to enhance operational efficiencies.

The application of fresh water substitutes can be extended to rig and other rigless activities such as CT cleanout, milling, descaling interventions.

Nomenclature

NF	nano-filtered
NFSW	nano-filtered seawater
ft	feet
gal	gallon
HCl	hydrochloric acid
MMSCFD	million standard cubic feet per day
PW	pond water
CT	coiled tubing
FWHP	flowing wellhead pressure
PI	productivity index
bbl	barrels
BPM	barrel per minute
DPW	distilled produced water
PPT	pounds per thousand gallons
GOSP	Gas Oil Separation Plant
EA	emulsified acid
PSI	pounds per square inch
HPHT	High Pressure High Temperature
GEOC	geo-chemical
WH	wellhead
Max	maximum
Avg	average
ISIP	instantaneous shut-in wellhead pressure
cP	centipoise
lb	pound